1. Identity

Document	STAR-PKG-STAR-CDM-EN · v2.2 · 2026-11-02
Package	star-cdm
Version	0.2.2
Capability	star.ssa.cdm
Maturity	Beta (VERIFIED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_cdm

2. Licence and obligations

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3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_cdm-0.2.2-py3-none-any.whl
```

FILE	SHA-256
star_cdm-0.2.2-py3-none-any.whl	ffc17253d27022b2e01af9d16047fffb7a65635 6f4f3d2392a7b3e27ce720610
star_cdm.py	101363f1c48f3a735efa1e1ef3670b58eb75ad1 d698b174bd3213b0cc8a5b8a2
star_cdm-0.2.2-source.zip	cf583ec86e3b3b63d9c8cbd111626296eab9bcd a48d6ba08df9c4f4b112f9bc9

5. What it does, and its limits

Strict parser for CCSDS 508.0-B-1 Conjunction Data Messages in KVN. Standard library only.

XML CDM, CDM 2.0 (508.0-B-2 draft/updates), collision probability computation, any operational use.

- `parse_cdm(text)` → `{"header", "object1", "object2", "warnings"}`: dates as datetimes (CCSDS 502.0 formats only), numbers as floats, units checked against the standard, exactly OBJECT1 then OBJECT2, all obligatory keywords (the set of the standard's example 3.6.2).
- Every malformation raises `CdmFormatError` (unknown keyword, missing obligatory keyword, wrong unit, duplicate, non-numeric, malformed date, wrong object count). Typographic minus signs (U+2212) are normalised and REPORTED.
- `covariance_rtn(obj)` → symmetric 6×6; `consistency_report(cdm)` compares MISS_DISTANCE with the state vectors and RELATIVE_POSITION_R/T/N and flags inconsistencies without editing values.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_cdm_0_2_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-cdm}: Strict CCSDS 508.0-B-1 Conjunction Data Message (KVN) parser with
consistency report},
  version = {0.2.2},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_cdm/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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